

ISHTAR TERRA

Sacajawea Patera



TYPE Caldera
AGE Under 500 million years
DIAMETER 145 miles (233 km)

Sacajawea Patera is an elliptically shaped volcanic caldera on Lakshmi Planum. It is thought to have formed when a large underground chamber was drained of magma and then collapsed. The resulting caldera then sagged. The depression is about 0.75 miles (1.2 km) deep and is enclosed by a zone of concentric troughs and scarps that extend up to 60 miles (100 km) in length and are 0.3–2.5 miles (0.5–4 km) apart. They are thought to have formed as the caldera

sagged. Sacajawea was a Shoshoni Indian woman, born in 1786, who worked as an interpreter.

SAG-CALDERA
Bright linear scarps extend out from Sacajawea Patera's eastern edge.

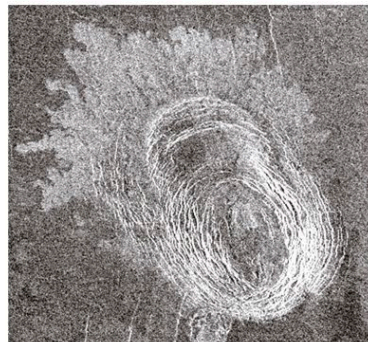
GUINEVERE PLANITIA

Sachs Patera



TYPE Caldera
AGE Under 500 million years
LENGTH 40 miles (65 km)

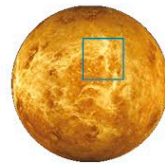
Sachs Patera is about 420 ft (130 m) deep and is surrounded by scarps spaced 1–3 miles (2–5 km) apart. A second, separately produced arc-shaped set of scarps lies to the north (top in the image below) of the main caldera. Solidified lava flows extend 6–16 miles (10–25 km) to the north and northwest of those scarps.



SCARPS AROUND SACHS PATERA

BETA REGIO

Beta Regio



TYPE Volcanic highland
AGE Under 500 million years
LENGTH 1,781 miles (2,869 km)

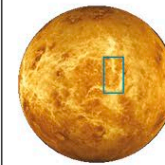
Beta Regio is a large highland region dominated by Rhea Mons and Theia Mons. Rhea, which lies 500 miles (800 km) to the north of Thea, was originally thought to be a volcano, but Magellan data revealed it to be an uplifted massif cut through by a rift valley, the Devana Chasma (right). Theia Mons is a volcano superimposed onto the rift.



RHEA AND THEIA MONS

BETA REGIO

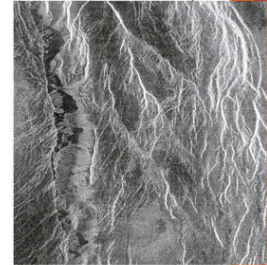
Devana Chasma



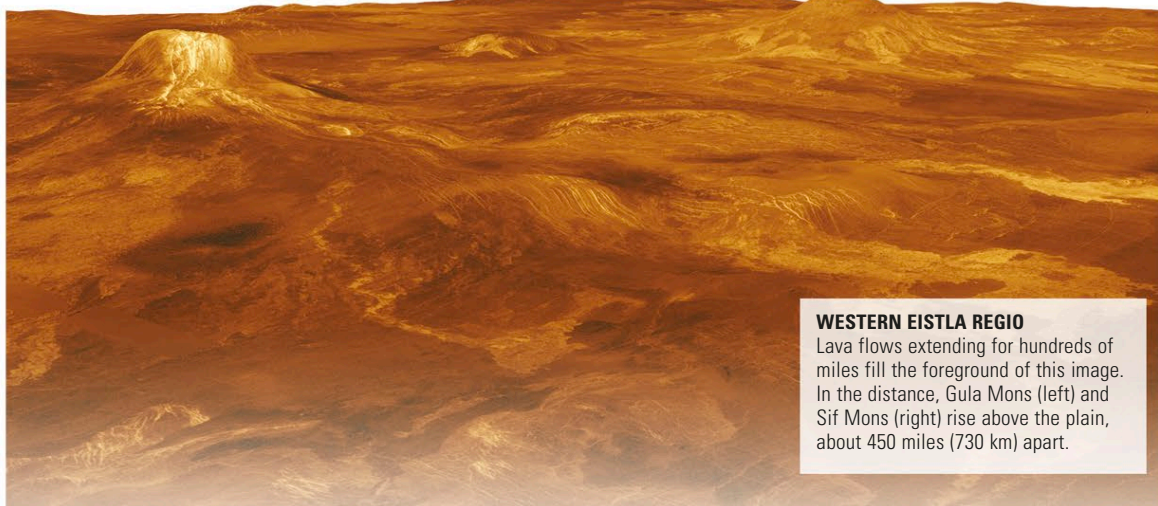
TYPE Fault
AGE Under 500 million years
LENGTH 2,860 miles (4,600 km)

Devana Chasma is a large fault that cuts through Beta Regio (left). This major rift valley extends in a north–south direction and was produced as the planet's crust pulled apart and the surface sank to form a valley floor with steep sides.

It is similar to the Great Rift Valley on Earth (see p.130). Devana Chasma slices through Rhea Mons and Theia Mons. The fault is over 1.2 miles (2 km) deep and about 50 miles (80 km) wide near Rhea Mons. Elsewhere it is broader, as much as 150 miles (240 km) wide. To the south of Theia Mons, it continues to the highland region Phoebe Regio and reaches depths of 3.7 miles (6 km). Faults and grabens cut through and fan out from parts of the rift valley.

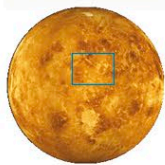


LANDSLIDE



EISTLA REGIO

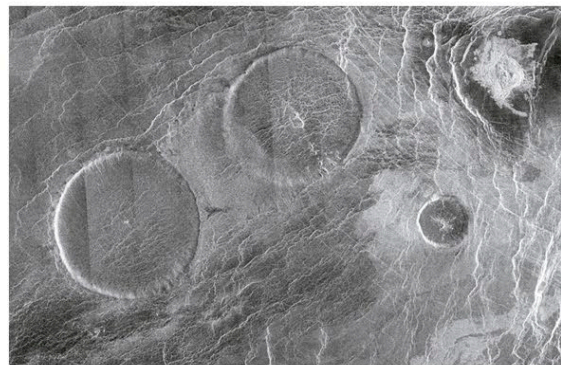
Eistla Regio



TYPE Volcanic highland
AGE Under 500 million years
LENGTH 4,977 miles (8,015 km)

Eistla Regio is one of Venus's smaller upland areas, which are located in the lower basin land separating the major highland areas. Eistla Regio lies in the equatorial region to the west of the major highland, Aphrodite Terra. It is a series of broad crustal rises, each of which is several thousand miles across. The landscape was seen for the first time in the 1980s, when data collected by the Pioneer Venus Orbiter was used to produce the first accurate

topographic map of Venus. Prominent features, such as the volcanoes Sif Mons and Gula Mons (right) and their lava flows, were clearly visible in the west of the region. Eistla Regio was also the first of the equatorial highlands imaged in the 1990s by Magellan, which revealed more detail



WESTERN EISTLA REGIO

Lava flows extending for hundreds of miles fill the foreground of this image. In the distance, Gula Mons (left) and Sif Mons (right) rise above the plain, about 450 miles (730 km) apart.

of the broad volcanic rises and rift zones. An unusual type of volcanic dome unique to Venus is found within Eistla Regio. The domes are circular, flat-topped mounds of lava and so are often referred to as pancake domes. It is believed that when the lava erupted through the surface, it

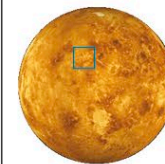
was highly viscous and so didn't flow freely. Cracks and pits in the domes are caused by cooling and withdrawal of lava.

PANCAKE DOMES

The two large, flat pancake domes are each about 37 miles (60 km) across and less than 0.6 miles (1 km) in height.

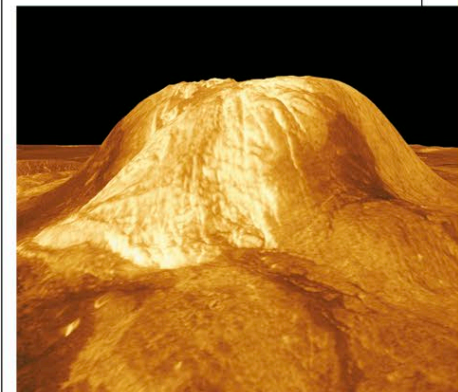
EISTLA REGIO

Gula Mons



TYPE Shield volcano
AGE Under 500 million years
HEIGHT 2 miles (3 km)

Gula Mons is the larger of the two volcanoes that dominate the highland rise of western Eistla Regio (left). At its widest, it measures about 250 miles (400 km) across. This shield volcano is encircled by hundreds of miles of lava flows. It does not have a caldera at its summit, but a fracture line 93 miles (150 km) long. The volcano is also at the center of an array of crustal fractures. A particularly prominent one, Guor Linea, is a rift system that extends for at least 600 miles (1,000 km) from the southeast flank.



SOUTHWEST SLOPES OF SUMMIT